THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–12) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A student claims that heavier objects fall faster. A good way to test it is to: 1

(a) wait for rain
(b) ask ten friends
(c) look at pictures
(d) drop objects of different masses from the same height

2. Which pair is correctly matched? 1

(a) pumpkin: creeper **(b)** spinach: tree **(c)** rose: herb **(d)** mango: climber

3. A child eating mostly polished rice and no other food may lack: 1

(a) carbohydrates **(b)** starch
(c) protein and vitamins **(d)** energy

4. A magnet is brought near an unknown rod. It is attracted at both ends. The rod is: 1

(a) plastic **(b)** surely non-magnetic
(c) surely a magnet **(d)** either a magnet or magnetic material

5. The length of a table is 1.2 m. In centimetres it is: 1

(a) 1200 cm **(b)** 120 cm **(c)** 0.12 cm **(d)** 12 cm

6. A student places a stone, a cork, an iron nail and a leaf in water. Which pair floats? 1

(a) cork and leaf **(b)** leaf and nail **(c)** nail and stone **(d)** stone and cork

7. A clinical thermometer shows 38.5 °C. The person has: 1

(a) no temperature **(b)** low temperature
(c) fever **(d)** normal temperature

8. A glass of water left in an open window for a week shows less water. This is because of: 1

(a) melting **(b)** evaporation **(c)** condensation **(d)** freezing

9. A mixture contains chalk powder, salt and iron filings. The first method to use is: 1

(a) filtration **(b)** sieving **(c)** evaporation **(d)** magnet

10. A student found a dry seed in a drawer. Which test shows that it is living? 1

(a) it is round **(b)** it is brown
(c) it sprouts when given water **(d)** it is light

11. Which of the following is a renewable but exhaustible if misused resource? 1

(a) petroleum **(b)** coal **(c)** natural gas **(d)** forests

12. Which of the following correctly lists planets in order from the Sun? 1

(a) Mars, Earth, Venus, Mercury **(b)** Mercury, Venus, Earth, Mars
(c) Earth, Mars, Venus, Mercury **(d)** Venus, Mercury, Earth, Mars

13. Priya's plants kept in a dark cupboard became pale. The conclusion is: 1

(a) soil is not needed **(b)** plants need light for healthy growth
(c) plants like cupboards **(d)** water is harmful

14. Which of these animals is both a land and water animal? 1

(a) goat **(b)** horse **(c)** camel **(d)** crocodile

15. Which should be eaten more in a daily diet? 1

(a) sweet drinks **(b)** fried snacks
(c) sweets **(d)** vegetables and fruits

16. Which of the following correctly lists only magnetic materials? 1

(a) copper, silver, gold **(b)** water, oil, air
(c) iron, nickel, cobalt **(d)** wood, rubber, glass

17. Which motion is both periodic and linear (back and forth along a line)? 1

(a) a falling stone **(b)** a mass vibrating on a spring
(c) a spinning top **(d)** a wheel

18. Why does a stone sink in water but a boat made of iron float? 1

(a) iron is lighter than stone **(b)** water pushes iron
(c) boats have no weight **(d)** the boat's shape holds a lot of air

19. Why does the reading of a clinical thermometer stay after it is removed from the mouth? 1

(a) because of the glass **(b)** because of the bulb
(c) because of the scale **(d)** because of the kink

20. Cloud formation requires: 1

(a) evaporation and condensation **(b)** freezing and melting
(c) boiling only **(d)** melting and freezing

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Why is it important to change only one variable in an experiment? 2

22. Why are cactus leaves modified into spines? 2

23. A child eats only rice and potatoes daily. Which nutrients are missing? 2

24. A student finds that a rod attracts both poles of a magnet. Can the rod be a magnet? Explain. 2

25. Convert 3.75 km into metres and centimetres. 2

26. Why do boats made of iron float while iron nails sink? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Design an experiment to test the hypothesis 'Seeds sprout faster in warm places'. 3

28. Match and explain: lotus, cactus, pine with water, desert and mountain. 3

29. A student says 'Junk food saves time and so it is good'. How would you reply? 3

30. A student has a mix of iron filings, sand and salt. How to separate all three? 3

31. A ribbon 2.4 m long is cut into pieces of 15 cm. Find the number of pieces and the leftover length. 3

OR

You are given a mixture of sand, iron filings and sugar. Describe how to separate them using properties. 3

32. A student reads a thermometer as 70 °C for hot water and 25 °C for the room. Describe what will happen to the water temperature in 2 hours. 3

OR

Describe the water cycle and name its stages. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Plan an investigation to find whether the length of a pendulum affects the time of its swing. Mention the variables. 5

34. Compare the adaptations of a camel, a fish, a bird and a polar bear for their habitats. 5

35. Prepare a plan to correct the diet of a child who eats chips, soda and noodles daily. 5

36. Describe an activity to show the repulsion between like poles and attraction between unlike poles and write the conclusion. 5

OR

Explain how you would measure the circumference of a cylindrical glass and its diameter, and verify the ratio. 5

37. Plan an investigation to find the best material for making a thermos flask lid among wood, plastic, steel and cork. 5

OR

A student heats water in a beaker and records the temperature every 2 minutes: 30, 38, 46, 54, 62 °C. Find the rise per 2 minutes and the likely temperature after 10 minutes. 5

38. Describe an experiment to show the effect of surface area, temperature and wind on evaporation. 5

OR

Plan the separation of a mixture containing sand, sugar, iron filings and water-insoluble sawdust. 5

END OF QUESTION PAPER — Total: 80 Marks

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